

Point Spatio-Temporal Transformer Networks for Point Cloud Video Modeling

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Abstract—Due to the inherent unorderedliness and irregularity of point cloud, points emerge inconsistently across different frames in a point cloud video. To capture the dynamics in point cloud videos, tracking points and limiting temporal modeling range are usually employed to preserve spatio-temporal structure. However, as points may flow in and out across frames, computing accurate point trajectories is extremely difficult, especially for long videos. Moreover, when points move fast, even in a small temporal window, points may still escape from a region. Besides, using the same temporal range for different motions may not accurately capture the temporal structure. In this paper, we propose a Point Spatio-Temporal Transformer (PST-Transformer). To preserve the spatio-temporal structure, PST-Transformer adaptively searches related or similar points across the entire video by performing self-attention on point features. Moreover, our PST-Transformer is equipped with an ability to encode spatio-temporal structure. Because point coordinates are irregular and unordered but point timestamps exhibit regularities and order, the spatio-temporal encoding is decoupled to reduce the impact of the spatial irregularity on the temporal modeling. By properly preserving and encoding spatio-temporal structure, our PST-Transformer effectively models point cloud videos and shows superior performance on 3D action recognition and 4D semantic segmentation.

Index Terms—Action recognition, point cloud, semantic segmentation, spatio-temporal modeling, video analysis

1 INTRODUCTION

UNDERSTANDING the dynamics of the 3D environment is important for intelligent agents to interact with the real world we live in. Benefiting from depth sensors, intelligent agents can directly obtain the dynamic visual information of the 3D world via point cloud videos or sequences. However, because each point cloud frame is unordered and irregular, points emerge inconsistently across different frames, making it challenging to model the spatio-temporal structure. Therefore, existing grid-based methods [4], [15], [43] on conventional RGB video processing cannot be directly employed to model point cloud videos. Although we can convert point cloud videos to 3D voxel sequences and then apply grid-based 4D convolutions, directly performing convolutions on voxel sequences requires a large amount of computation and special engineering efforts, e.g., sparse convolution [5]. Furthermore, quantization errors are inevitable during voxelization, which may hinder applications that require precise point positions.

To preserve the spatio-temporal structure in raw point cloud videos, MeteorNet [27] applies point tracking to find

the trajectory of each point in a video, as shown in Fig. 1a. However, because points may flow in and out across frames, accurately tracking points is extremely difficult, especially for long videos. To avoid point tracking, PSTNet [12] limits the temporal modeling range by observing only a few frames (temporal window) for each local area, assuming that a point would not escape from the area in a short period of time, as shown in Fig. 1b. However, when frame sampling frequency is low or objects move fast, points may still escape from local areas. Moreover, the optimal temporal window size for different motions is usually different. Using the same temporal modeling range for all motions may not properly capture the dynamics. Despite the disadvantages in preserving spatio-temporal structure, MeteorNet and PSTNet have an ability to encode spatio-temporal structure based on spatial displacements and temporal differences.

In this paper, we propose a Point Spatio-Temporal Transformer (PST-Transformer) for point cloud video modeling. To preserve spatio-temporal structure, our PST-Transformer employs a video-level self-attention to adaptively search related or similar points across entire videos. In this way, related points are merged with attention weights, instead of explicitly tracking points or limiting temporal modeling range. Moreover, different from the previous P4Transformer work [11], as shown in Fig. 1c, which mainly perform weighted sum on features, our PST-Transformer is equipped with an ability to encode spatio-temporal structure. As shown in Fig. 1d, to encode the spatio-temporal structure of a reference point, the features in a video are encoded based on the spatial-temporal displacements from the reference to all the points in the video. Then, the reference feature is updated as the weighted sum on the encoded features, in which the unrelated points are excluded. In this

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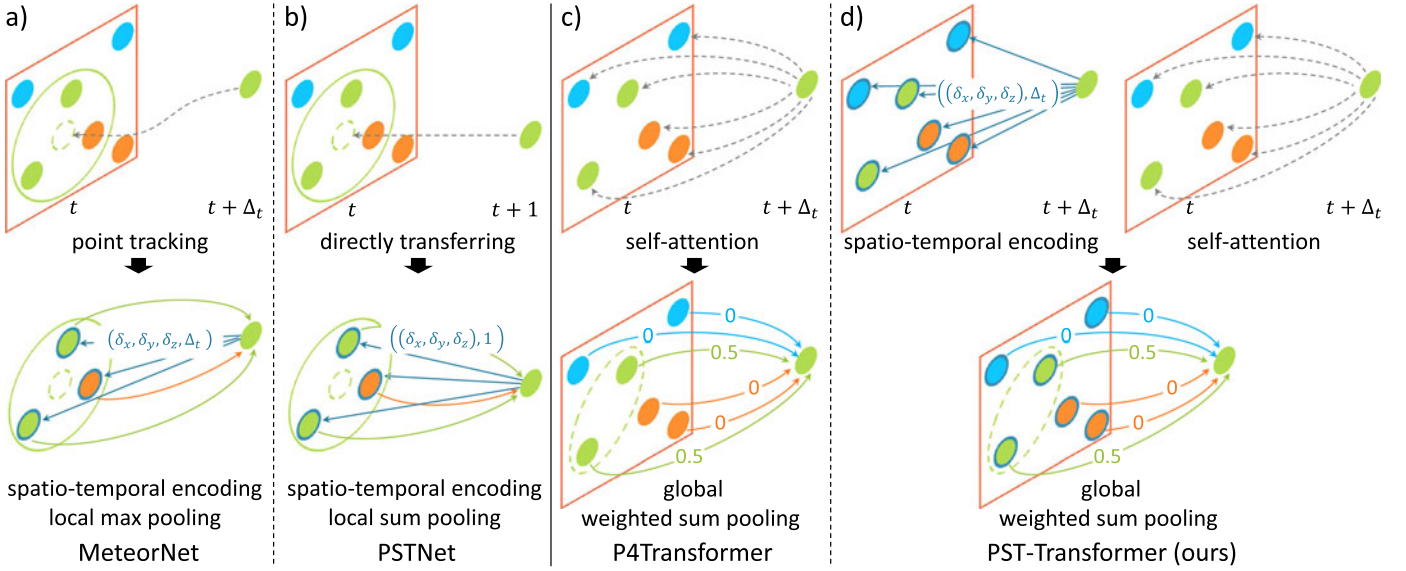


Fig. 1. Illustration of different point-based methods for spatio-temporal modeling in point cloud videos. **a)** MeteorNet [27] employs point tracking to preserve spatio-temporal structure. **b)** PSTNet [12] limits temporal modeling range (e.g., 1 neighboring frame) to preserve spatio-temporal structure. **c)** P4Transformer [11] is able to adaptively search related points across entire videos based on self-attention and global weighted sum pooling. However, it does not encode the spatio-temporal structure like MeteorNet or PSTNet. **d)** The proposed PST-Transformer is equipped with an ability to encode spatio-temporal structure based on the spatial displacement $(\delta_x, \delta_y, \delta_z)$ and temporal difference Δ_t .

way, the spatio-temporal structure is preserved and encoded. Because point coordinates are irregular and unordered but point timestamps exhibit regularities and order, we follow [12] to decouple the spatio-temporal encoding to reduce the impact of the spatial irregularity on the temporal modeling. Extensive experiments on widely-used 3D action recognition and 4D semantic segmentation datasets demonstrate the superiority of PST-Transformer in modeling point clouds videos. The contributions of this paper are threefold:

- To preserve spatio-temporal structure, we propose a Point Spatio-Temporal Transformer (PST-Transformer) to adaptively search related or similar points for raw point cloud video modeling.
- Besides structure preservation, our PST-Transformer has an additional ability to encode the spatio-temporal structure of raw point cloud videos in a decoupled manner.
- Extensive experiments on four datasets show that the proposed PST-Transformer effectively improves the accuracy of 3D action recognition and 4D semantic segmentation.

This paper is an extension of our previous work P4Transformer [11]. P4Transformer mainly employs the original Transformer [8], [46] to preserve spatio-temporal structure but does not encode spatio-temporal structure. In this paper, inspired by MeteorNet [27] and PSTNet [12], we integrate spatio-temporal structure encoding into Transformer, resulting in the proposed PST-Transformer.

2 RELATED WORK

Deep Learning on Grid-based RGB Videos. Conventional grid-based RGB videos are essentially image sequences. To model the spatio-temporal structure in videos, two-stream

convolutional neural networks [41], [49] employ a spatial stream to embed appearance and an optical flow stream to capture motion. As video is a kind of sequence, recurrent neural networks [9], [29] are also used to capture the temporal dependencies of videos. In addition, by stacking multiple 2D frames into a 3D tensor, 3D convolutional neural networks [4], [15], [43], [44] directly learn spatio-temporal representations for videos. However, these methods are applied to grid-based data and cannot be directly used for processing raw point cloud videos. Besides, interpretable video or action reasoning methods [13], [60] are proposed by explicitly parsing changes in videos.

Dynamic Point Cloud Modeling. Learning from dynamic point clouds is fairly new but important for intelligent agents to interact with the dynamic 3D world. Three main types of methods have been explored. The first type projects 3D point cloud frames to 2D images and then extracts features in the same way as RGB video modeling via 3D convolutions [28]. Due to the projected dimensions, these methods distort the real 3D shape [52] and may increase the challenge for networks to understand 3D structures. The second type is to convert a 3D point cloud video into 3D voxel sequences and then apply 4D grid-based convolutions to these voxels to capture the dynamics [5]. However, as points are usually sparse, directly performing convolutions on the entire space along the time dimension is computationally inefficient. Special engineering efforts, e.g., sparse convolution, are usually needed. Moreover, quantization errors are inevitable during voxelization, which may restrict applications that require precise measurement of scene geometry. The last type is point-based and directly models raw point cloud videos. For example, PointRNN [10] proposes a point-based method to recurrently capture the spatio-temporal structure in point cloud videos. MeteorNet [27] extends 3D points to 4D points by appending a temporal dimension and then extends PointNet++ [34] to process 4D

points. PSTNet [12] proposes a point spatio-temporal convolution to model raw point cloud videos in a hierarchical manner. P4Transformer [11] employs Transformer to avoid point tracking but does not encode the spatio-temporal structure. Besides, 3DV [52] combines voxel-based and point-based modeling by first integrating 3D motion information into a regular compact voxel set and then applying PointNet++ to extract representations from the set for 3D action recognition via temporal rank pooling. In this paper, we propose a new point-based method, i.e., PST-Transformer, to adaptively search related or similar regions while encoding spatio-temporal structure for point cloud modeling.

Dynamic point cloud modeling is also employed and studied in other scenarios. For example, Niemeyer *et al.* [30] learned a temporal and spatial vector field in which every point is assigned with a motion vector of space and time for 4D reconstruction. Prantl *et al.* [33] learned stable and temporal **coherent** feature spaces for point-based super-resolution. CaSPR [35] learns to encode spatio-temporal changes of object shapes from dynamic point clouds for reconstruction and camera pose estimation. Besides, scene flow estimation [14], [26] can be seen as computing the motion between two point cloud frames.

Transformer Networks. Impressive progress has been made on natural language processing due to the success of Transformer networks [6], [7], [46], [56]. In computer vision, the community has used self-attention or Transformer to model images in a non-local manner [3], [8], [17], [51], [58]. In particular, Zhao *et al.* proposed a PointTransformer [59] to model static point clouds. Inspired by these methods, we propose a PST-Transformer to avoid point tracking when processing raw point cloud videos. Moreover, our approach integrates spatio-temporal structure encoding into Transformer for the representation learning of point cloud videos.

3 POINT SPATIO-TEMPORAL TRANSFORMER NETWORK

In this section, we first briefly review the original Transformer and discuss its potential limitation for structure modeling. Then, we describe the proposed Point Spatio-Temporal Transformer (PST-Transformer) in detail. Finally, we incorporate our PST-Transformer into deep neural networks to address two dynamic point cloud tasks, i.e., 3D action recognition and 4D semantic segmentation.

3.1 Preliminary: Transformer

The goal of Transformer [8], [46] is to merge related regions based on their similarities so that each position has a larger receptive field to collect more information from its neighbors or related positions. Suppose $F \in \mathbb{R}^{C \times M}$ is the input feature map, where M is the length of the map and C is the number of feature channels. Transformer first performs two individual feature encodings on the feature map to generate queries $Q \in \mathbb{R}^{C' \times M}$ and keys $K \in \mathbb{R}^{C' \times M}$, where C' is the query or key dimension. Then, the scaled dot-product attention $A \in \mathbb{R}^{M \times M}$ is computed for each query and a softmax function is applied to obtain the attention weights between every two positions,

$$Q = W_q \cdot F, \quad K = W_k \cdot F, \quad A = \frac{Q^T \cdot K}{\sqrt{C'}}, \quad \alpha_{ij} = \frac{e^{A_{ij}}}{\sum_{m=1}^M e^{A_{im}}}, \quad (1)$$

where $\cdot$ is the matrix multiplication and $W_q, W_k \in \mathbb{R}^{C' \times C}$. The α_{ij} denotes the attention weight of the j th position on the i th position. Third, Transformer employs another feature encoding on the input feature map to generate values $V \in \mathbb{R}^{C'' \times M}$, where C'' is the value dimension. Finally, the output is computed as a weighted sum of the values,

$$V = W_v \cdot F, \quad O_i = \sum_{j=1}^M \alpha_{ij} V_j, \quad (2)$$

where $W_v \in \mathbb{R}^{C'' \times C}$ and $O \in \mathbb{R}^{C'' \times M}$.

Based on self-attention and global weighted sum, Transformer is able to adaptively search related regions, thus more flexible than local methods in structure preservation. However, it does not encode the structure like MeteorNet [27] or PSTNet [12] that learns a feature encoding for every spatio-temporal direction and distance.

3.2 PST-Transformer for Point Cloud Video Modeling

A point cloud video can be represented by a sequence of 3D point coordinates $P \in \mathbb{R}^{3 \times L \times N}$, where L is the video length and N is the number of points per frame. Moreover, when point colors or other attributes are available, they are often used as additional features $F \in \mathbb{R}^{C \times L \times N}$ associated with the 3D points to enhance the discriminativeness of point clouds, where C denotes the number of feature channels. Given a point cloud video, the proposed PST-Transformer will generate new point features $F' \in \mathbb{R}^{C'' \times L \times N}$ based on the input point coordinates and features, where C'' represents the new number of feature channels.

3.2.1 Spatio-Temporal Structure Preservation

To avoid tracking points or limiting temporal modeling range, we exploit self-attention to preserve the spatio-temporal structure for point cloud video modeling. Similar to the original Transformer, our PST-Transformer first performs two individual feature encodings to generate queries $Q \in \mathbb{R}^{C' \times L \times N}$ and keys $K \in \mathbb{R}^{C' \times L \times N}$ and then computes the scaled dot-product attention $A \in \mathbb{R}^{L \times L \times N}$ for all points. To search similar or related points across the entire point cloud video, we propose a video-level self-attention, in which softmax is applied to the entire video for each point,

$$\alpha_{pp'} = \frac{e^{A_{pp'}}}{\sum_{t'=1}^L \sum_{p'' \in P_{t''}} e^{A_{pp''}}} \quad (3)$$

where $\alpha_{pp'}$ denotes the attention weight of point p at the (x, y, z) position in the t th frame on point p' at the (x', y', z') position in the t' th frame. The $P_{t''}$ denotes all the points in the t'' th frame and p'' denotes the point at the (x'', y'', z'') position in the t'' th frame. In this way, the related or similar points will be selected with high attention weights.

For the video-level self-attention, the sum of all the weights on a point is 1, assuming that the point may not appear in all frames. Another option is the so-called frame-

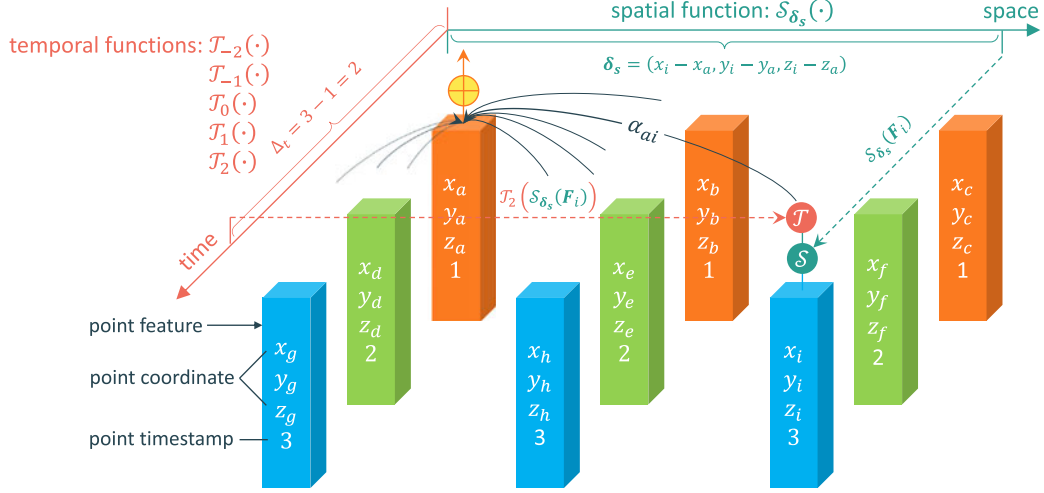


Fig. 2. Illustration of the proposed PST-Transformer. The input contains $L = 3$ frames, with $N = 3$ points per frame. PST-Transformer is updating the feature of point a in the 1st frame based on all the points in the video. When modeling the spatio-temporal structure from point a to point i in the 3rd frame, PST-Transformer first employs a video-level self-attention to achieve the attention weight α_{ai} , which indicates how much point i is similar or related to point a . Then, to encode the spatio-temporal structure from point a to point i , PST-Transformer applies a spatial structure encoding function $S_{ss}(\cdot)$ and a temporal structure encoding function $T_2(\cdot)$ on the feature of point i based on the spatial displacement δ_s and temporal difference $\Delta_t = 2$, respectively. Finally, the encoded feature of point i is added into the new feature of point a based on the attention weight α_{ai} .

level self-attention, which applies softmax to each frame individually, i.e.,

$$\alpha_{pp'} = \frac{e^{A_{pp'}}}{\sum_{p'' \in P_t} e^{A_{pp''}}}, \quad (4)$$

so that the sum of the weights in each frame is 1. Similar to point tracking, the frame-level self-attention assumes that each query point appears in every frame. However, because points may flow in and out across frames and disappearing points may appear again, point trajectories are usually not continuous. Therefore, the frame-level self-attention is not suitable for preserving the spatio-temporal structure of point cloud videos.

3.2.2 Spatio-Temporal Structure Encoding

Inspired by MeteorNet [27] and PSTNet [12], we propose to integrate the ability of spatio-temporal structure encoding into PST-Transformer for point cloud video modeling. Let $\delta s = (x' - x, y' - y, z' - z)$ and $\Delta t = t' - t$ denote the spatial displacement and temporal difference from point p to point p' , respectively. Our PST-Transformer is designed to learn a spatio-temporal structure encoding function $\mathcal{F}_{(\delta s, \Delta t)}(\cdot)$ for each spatio-temporal displacement $(\delta s, \Delta t)$ to generate values,

$$\mathbf{V}_{pp'} = \mathcal{F}_{(\delta s, \Delta t)}(F_{p'}), \quad (5)$$

where $F_{p'} \in \mathbb{R}^{C \times 1}$ denotes the feature of point p' . The $\mathbf{V} \in \mathbb{R}^{C'' \times LN \times LN}$ is the values that encodes the spatio-temporal structure for any two points in a video. As point cloud videos are spatially irregular and unordered but temporally ordered, we should reduce the impact of the spatial irregularity of points on the temporal modeling. Moreover, because space and time are orthogonal and independent of each other, and the scales of spatial displacements and temporal differences may be incompatible, the spatial and temporal structure should be encoded separately. Therefore, we

follow [12] to decouple $\mathcal{F}_{(\delta s, \Delta t)}(\cdot)$ as follows,

$$\mathbf{V}_{pp'} = T_{\Delta t}(S_{\delta s}(F_{p'})), \quad (6)$$

where $S_{\delta s}(\cdot)$ is a spatial structure encoding function and $T_{\Delta t}(\cdot)$ is a temporal structure encoding function. Note that, because point timestamps are regular and discrete, for a video with length L , there are $2L - 1$ temporal functions, i.e., $T_{-(L-1)}(\cdot), \dots, T_0(\cdot), \dots, T_{(L-1)}(\cdot)$. Finally, the global weighed sum is performed on the values $\mathbf{V}$ to obtain the new feature for point p ,

$$O_p = \sum_{t'=1}^L \sum_{p' \in P_{t'}} \alpha_{pp'} \mathbf{V}_{pp'}, \quad (7)$$

where $O \in \mathbb{R}^{C'' \times LN}$.

Similar to the original Transformer [8], [46], our PST-Transformer also employs multi-head attention, ReLU activation functions, LayerNorms [1] and residual connections to enhance the modeling ability and generate the new point features $F' \in \mathbb{R}^{C'' \times LN}$, as shown in Fig. 3.

3.2.3 Implementation of $S_{\delta s}(\cdot)$ and $T_{\Delta t}(\cdot)$

There can be many ways to implement the $S_{\delta s}(\cdot)$ and $T_{\Delta t}(\cdot)$ functions. For the spatial structure encoding function $S_{\delta s}(\cdot)$, we can employ continuous convolutions [12], [42], [45], [53], or displacement encoding methods [27], [59]. In this paper, we employ both of the two types of methods to implement $S_{\delta s}(\cdot)$, respectively,

$$\begin{aligned} \text{convolution: } S_{\delta s}(F_{p'}) &= f(\delta s; \theta) \cdot F_{p'}, \\ \text{concatenation: } S_{\delta s}(F_{p'}) &= W_s \cdot [F_{p'}; \delta s^T], \end{aligned} \quad (8)$$

where $f: \mathbb{R}^{1 \times 3} \rightarrow \mathbb{R}^{C'' \times C}$ is a function of δs parametrized by θ to generate different convolution kernels according to spatial displacements, $[\cdot; \cdot]$ denotes concatenation and $W_s \in \mathbb{R}^{C'' \times (C+3)}$. The f function is implemented as the same way as PSTNet [12]. The concatenation method can be seen as

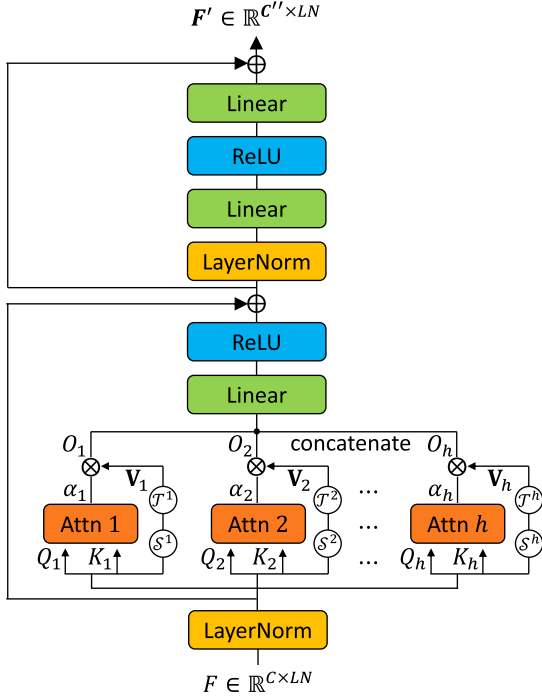


Fig. 3. Illustration of PST-Transformer with multi-head attention, ReLUs, LayerNorms and residual connections, where h is the number of heads.

the sum of a feature encoding and a relative position encoding in PointTransformer [59]. By default, we employ the displacement concatenation method in experiments.

For the temporal encoding function $\mathcal{T}_{\Delta t}(\cdot)$, we implement it as follows,

$$\mathcal{T}_{\Delta t}(\mathcal{S}_{\delta s}(F_{p'})) = w_{\Delta t} \mathcal{S}_{\delta s}(F_{p'}), \quad (9)$$

where $w_{\Delta t}$ is a learnable scalar parameter for the temporal difference Δt . An illustration of the proposed PST-Transformer is shown in Fig. 2.

3.3 Architectures

To evaluate the ability to model point cloud videos, we incorporate our PST-Transformer into deep networks for 3D action recognition and 4D semantic segmentation.

3.3.1 Spatio-Temporal Local Area Encoding

Because a point cloud video may contain a large number of points, directly perform self-attention on all points requires a large amount of computation and running memory. Moreover, point clouds are not always associated with point features. Without point features, it is difficult to perform self-attention. To address this problem, we follow our previous P4Transformer [11] work to first sample a few points to represent spatio-temporal local areas and then embed the local areas as the features of these points, as shown in Fig. 4a.

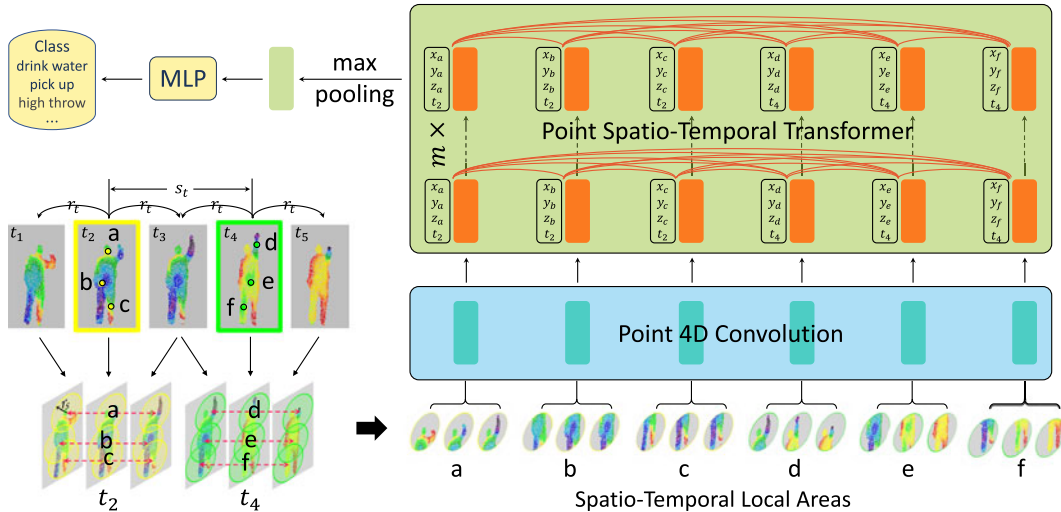
3.3.2 PST-Transformer for 3D Action Recognition

Action recognition is a fundamental task for video modeling, which can be seen as a video-level classification problem. As shown in Fig. 4a, a PST-Transformer with multiple (m) layers is used to capture appearance and motion information across all the spatio-temporal local features. Then, a max pooling merges the transformed local features to a single global one. Finally, a multilayer perceptron (MLP) layer converts the global feature to action predictions.

3.3.3 PST-Transformer for 4D Semantic Segmentation

Semantic segmentation can be seen as a point-wise classification task. We apply 4D semantic segmentation to the self-

a) 3D Action Recognition



b) 4D Semantic Segmentation

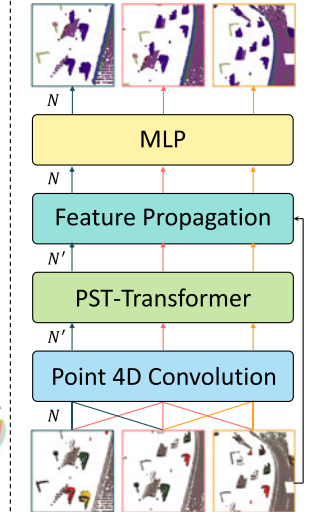


Fig. 4. Illustration of PST-Transformer networks. **a)** 3D action recognition. The input contains 5 frames, with N points per frame. (1) Based on temporal radius $r_t = 1$ and temporal stride $s_t = 2$, the t_2 and t_4 frames are selected. Suppose the spatial subsampling rate $s_s = N/3$. Then, $N/s_s = 3$ points are sampled for each selected frame. Based on the sampled points, spatial radius r_s and temporal radius r_t , spatio-temporal local areas are constructed. (2) Each spatio-temporal local area is encoded to a vector by point 4D convolution [11] as the region feature. (3) Our PST-Transformer updates local features by collecting more information from similar or related areas. (4) A max pooling merges local features to a global feature, which is then mapped to action predictions by a multilayer perceptron (MLP). **b)** 4D semantic segmentation. Because points are subsampled, after PST-Transformer, we employ a feature propagation operation [34] to recover the sampled N' points to the original N points by interpolating features. Finally, an MLP maps interpolated features to point predictions.

TABLE 1
Efficiency and Accuracy Comparison on the MSR-Action3D Dataset [20] With 24 Frames

Method	# Params	Memory	Time (ms)	Acc (%)
MeteorNet [27]	17.6 M	16,993 M	80.11	88.50
PSTNet [12]	8.4 M	21,367 M	63.88	91.20
P4Transformer [11]	44.1 M	9,971 M	25.18	90.94
PST-Transformer (small)	0.3 M	3,911 M	5.78	93.03
PST-Transformer	44.2 M	11,039 M	69.37	93.73

driving scenario. Different from 3D action recognition, which requires modeling long temporal dependencies for action reasoning, the self-driving scenario usually exploits only a few frames for each video, e.g., 3 frames, as scenes change dramatically during driving. Therefore, we do not subsample frames. After PST-Transformer, because point 4D convolutions subsample points, we add feature propagation layers [34] to interpolate point features. After the last feature interpolation layer, we add an MLP to convert point features to point predictions. The PST-Transformer framework for 4D semantic segmentation is shown in Fig. 4b.

4 EXPERIMENTS

4.1 3D Action Recognition

For 3D action recognition, we follow [11], [12], [27], [52] to sample 2,048 points for each frame. Point cloud videos are split into multiple clips (with a fixed number of frames) as inputs. For training, video-level labels are used as clip-level labels. For evaluation, the mean of the clip-level predicted probabilities is used as the video-level prediction. Point colors are not used.

Following [11], the temporal radius of point 4D convolution is set to 1 to capture temporal local correlation and the temporal stride is set to 2 to subsample frames. The spatial subsampling rate is set to 32. In this way, for a point video with 24 frames, PST-Transformer will process $\frac{2048}{32} \times \frac{24}{2} = 768$ points. We train our models for 50 epochs with the SGD optimizer. Learning rate is set to 0.01, and decays with a rate of 0.1 at the 20th epoch and the 30th epoch, respectively. We compare our PST-Transformer with skeleton-based, depth-based and point-based methods.

4.1.1 MSR-Action3D

The MSR-Action3D [20] dataset consists of 567 Kinect v1 depth videos, including 20 action categories and 23 K frames in total. The spatial radius of point 4D convolution is set to 0.3. The network contains 5 PST-Transformer layers, with 8 heads per layer. Batch size is set to 16. We use the same training/test split as previous works [11], [12], [27], [48].

I. Accuracy Comparison

The accuracy comparisons are reported in Table 3. Our method outperforms all the state-of-the-art methods, demonstrating the superiority of the proposed PST-Transformer on point cloud video feature extraction. Compared to our previous P4Transformer [11], PST-Transformer effectively improves the accuracy, showing the effectiveness of spatio-temporal structure encoding. For example, when using 24

TABLE 2
Comparison Between Convolution-Based and Concatenation-Based Spatial Structure Encoding Methods on the PST-Transformer (small) Model

Spatial encoding	24	32	40	48	56	64
Convolution	92.68	92.68	92.33	92.00	91.64	91.30
Concatenation	93.13	93.03	92.43	92.43	91.30	90.86

The 3D Action Recognition Experiment is Conducted on MSR-Action3D [20] With 24 Frames. Different Spatial Subsampling Rates are Evaluated

frames, PST-Transformer achieves an improvement of 2.79% compared to P4Transformer.

II. Efficiency Comparison

To investigate the running time, we conduct experiments using a single Nvidia Quadro RTX 6000 GPU. As shown in Table 1, compared to P4Transformer [11], the spatio-temporal structure encoding in PST-Transformer introduces only a few parameters.

Although P4Transformer and PST-Transformer use more parameters than MeteorNet and PSTNet, they do not consume too much running memory and time to achieve similar or better accuracy. This is because, after the point 4D convolution, P4Transformer and PST-Transformer require fewer points.

We also reduce the number of point feature channels to construct a small PST-Transformer model. Due to the spatio-temporal structure encoding ability, even the small model can outperform P4Transformer. To make the advantage of spatio-temporal structure encoding clear, suppose there are α actions that occur at β different 4D positions. P4Transformer injects 4D positions into point features and thus needs to model $\alpha \times \beta$ cases. In contrast, the proposed

TABLE 3
3D Action Recognition on MSR-Action3D [20] With Different Numbers of Frames

Method	Input	# Frames	Accuracy
Vieira <i>et al.</i> [47]	depth	20	78.20
Kläser <i>et al.</i> [16]	depth	18	81.43
Actionlet [48]	skeleton	all	88.21
PointNet++ [34]	point	1	61.61
MeteorNet [27]	point	4	78.11
PSTNet [12]	point	4	81.14
P4Transformer [11]	point	4	80.13
PST-Transformer (ours)	point	4	81.14
MeteorNet [27]	point	8	81.14
PSTNet [12]	point	8	83.50
P4Transformer [11]	point	8	83.17
PST-Transformer (ours)	point	8	83.97
MeteorNet [27]	point	12	86.53
PSTNet [12]	point	12	87.88
P4Transformer [11]	point	12	87.54
PST-Transformer (ours)	point	12	88.15
MeteorNet [27]	point	16	88.21
PSTNet [12]	point	16	89.90
P4Transformer [11]	point	16	89.56
PST-Transformer (ours)	point	16	91.98
MeteorNet [27]	point	24	88.50
PSTNet [12]	point	24	91.20
P4Transformer [11]	point	24	90.94
PST-Transformer (ours)	point	24	93.73

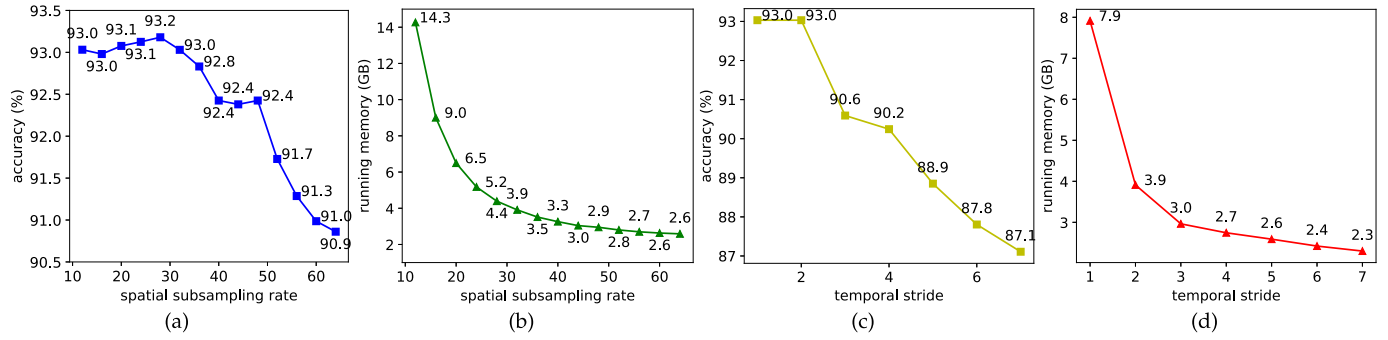


Fig. 5. Influence of spatial subsampling rate and temporal stride on the accuracy and running memory of the PST-Transformer (small) model. We conduct 3D action recognition on the MSR-Action3D [20] dataset with 24 frames.

PST-Transformer only needs to learn the α actions, thus requiring fewer parameters.

Finally, we investigate the influence of point subsampling on accuracy and running memory. In order to conduct experiments on videos with lowly subsampled points and frames, which may contain a large number of points, we employ the PST-Transformer (small) model. Recall that, the spatial sampling rate is set to 32 and temporal stride is set to 2 by default. The results are shown in Fig. 5. When the spatial subsampling rate is less than 32, the proposed model achieves similar accuracies, which indicates that the spatial subsampling rate of 32 is sufficient to capture the spatial structure for PST-Transformer. Using small spatial subsampling rates does not effectively improve accuracy but requires a large amount of running memory. When the temporal stride is greater than 2, the accuracy decreases. Although employing large temporal strides can reduce selected frames and save running memory, the model fails to capture the complete temporal structure, thus leading to inferior accuracies. Note that, because the temporal radius is set to 1, the point 4D convolution processes 3 frames each time, including 1 anchor frame and 2 neighboring frames. In this case, temporal stride 2 is sufficient to capture the complete temporal structure. Using temporal stride 1 is redundant and thus it is difficult to further improve accuracy.

III. Comparison of different spatial structure encoding methods.

As shown in Eq. (8), we employ two implements for spatial structure encoding, i.e., continuous convolution and displacement concatenation. To investigate the two implements, we conduct 3D action recognition on MSR-Action3D with different spatial subsampling rates. As shown in Table 2, the convolution and concatenation methods achieves similar accuracies. The difference is that, convolution achieves better accuracies on sparse points (large subsampling rates) while concatenation achieves better accuracies on dense points (small subsampling rates).

4.1.2 NTU RGB+D 60 and NTU RGB+D 120

The NTU RGB+D 60 [37] dataset is the second largest dataset for 3D action recognition. It consists of 56 K videos, with 60 action categories and 4 M frames in total. The videos are captured using Kinect v2, with 3 cameras and 40 subjects (performers). The dataset defines two types of evaluation, i.e., cross-subject and cross-view. The cross-subject evaluation

splits the 40 performers into training and test groups. Each group consists of 20 performers. The cross-view evaluation uses all the samples from camera 1 for testing and samples from cameras 2 and 3 for training.

The NTU RGB+D 120 [21] dataset is an extension of NTU RGB+D 60, which is the largest dataset for 3D action recognition. It consists of 114 K videos, with 120 action categories and 8 M frames in total. The videos are also captured by Kinect v2, with 106 performers and 32 collection setups (locations and backgrounds). Besides cross-subject evaluation, the dataset defines a new evaluation setting, i.e., cross-setup, where 16 setups are used for training, and the others are used for testing.

The spatial radius of point 4D convolution is set to 0.1. The network employs 5 PST-Transformer layers, with 8 heads per layer. Each clip contains 24 frames. Batch size is set to 32.

As shown in Table 4, the proposed PST-Transformer achieves three best accuracies among the four evaluation cases. Particularly, as indicated by the cross-setup evaluation on the largest NTU RGB+D 120 dataset, our PST-Transformer outperforms P4Transformer [11] by 1.1%, demonstrating the effectiveness of the spatio-temporal encoding.

4.1.3 What Does PST-Transformer Learn?

To investigate what PST-Transformer learns, we first visualize the attention weight and output in Fig. 6. The network is trained on MSR-Action3D. For the output, we visualize the point features after the first ReLU in Fig. 3 and squeeze the features to scalars via l_1 norm. Because the outputs of ReLU are greater than zero, large scalar values indicate high activations. As expected, PST-Transformer pays correct attention on a region across different frames, which is similar to tracking. Moreover, PST-Transformer outputs higher activations on moving areas, showing its ability to capture point dynamics for point cloud video modeling.

Then, we visualize the scalar parameters $\{w_{\Delta t}\}$ of temporal functions $\{T_{\Delta t}(\cdot)\}$ in Fig. 7. The network is trained on NTU RGB+D 120. For a better and clearer view, we only show 4 heads. Because each clip contains 24 frames and the point 4D convolution subsamples frames with a rate of 2, our PTS-Transformer takes $24/2 = 12$ frames as input. Therefore, $\Delta t \in \{-11, \dots, 0, \dots, 11\}$. As shown in the figure, PST-Transformer pays more attention on neighboring frames than others when modeling the current frame. More

TABLE 4
3D Action Recognition Accuracy (%) on NTU RGB+D 60 [37] and NTU RGB+D 120 [21]

Method	Input	NTU 60		NTU 120	
		Subject	View	Subject	Setup
SkeleMotion [2]	skeleton	69.6	80.1	67.7	66.9
GCA-LSTM [24]	skeleton	74.4	82.8	58.3	59.3
FSNet [22]	skeleton	-	-	59.9	62.4
Attention LSTM [23]	skeleton	77.1	85.1	61.2	63.3
Evolution Map [25]	skeleton	-	-	64.6	66.9
AGC-LSTM [40]	skeleton	89.2	95.0	-	-
AS-GCN [19]	skeleton	86.8	94.2	-	-
VA-fusion [57]	skeleton	89.4	95.0	-	-
2s-AGCN [39]	skeleton	88.5	95.1	-	-
DGNN [38]	skeleton	89.9	96.1	-	-
HON4D [32]	depth	30.6	7.3	-	-
SNV [55]	depth	31.8	13.6	-	-
HOG ² [31]	depth	32.2	22.3	-	-
Li <i>et al.</i> [18]	depth	68.1	83.4	-	-
Wang <i>et al.</i> [50]	depth	87.1	84.2	-	-
MVDI [54]	depth	84.6	87.3	-	-
NTU 120 Baseline [21]	depth	-	-	48.7	40.1
PointNet++ [34]	point	80.1	85.1	72.1	79.4
3DV [52]	voxel	84.5	95.4	76.9	92.5
3DV-PointNet++ [52]	voxel + point	88.8	96.3	82.4	93.5
PSTNet [12]	point	90.5	96.5	87.0	93.8
P4Transformer [11]	point	90.2	96.4	86.4	93.5
PST-Transformer (ours)	point	91.0	96.4	87.5	94.0

interestingly, parameters appear to follow Gaussian distributions with different standard deviations. This implies that different heads probably capture the temporal structure at different scales.

4.2 4D Semantic Segmentation

We perform 4D semantic segmentation on the Synthia 4D [5] dataset. Synthia 4D uses the Synthia dataset [36] to create 3D

videos, which includes 6 videos of driving scenarios. Following [11], [12], [27], we use the same training/validation/test split, with 19,888/815/1,886 frames, respectively, and conduct experiments on video clips with length of 3 frames, with 16,384 points per frame. The model contains four point 4D convolution layers, with the spatial subsampling rates of 4, 4, 4 and 2, respectively. In this way, our PST-Transformer takes $\frac{16384}{4 \times 4 \times 4 \times 2} \times 3 = 384$ points as input. The PST-Transformer

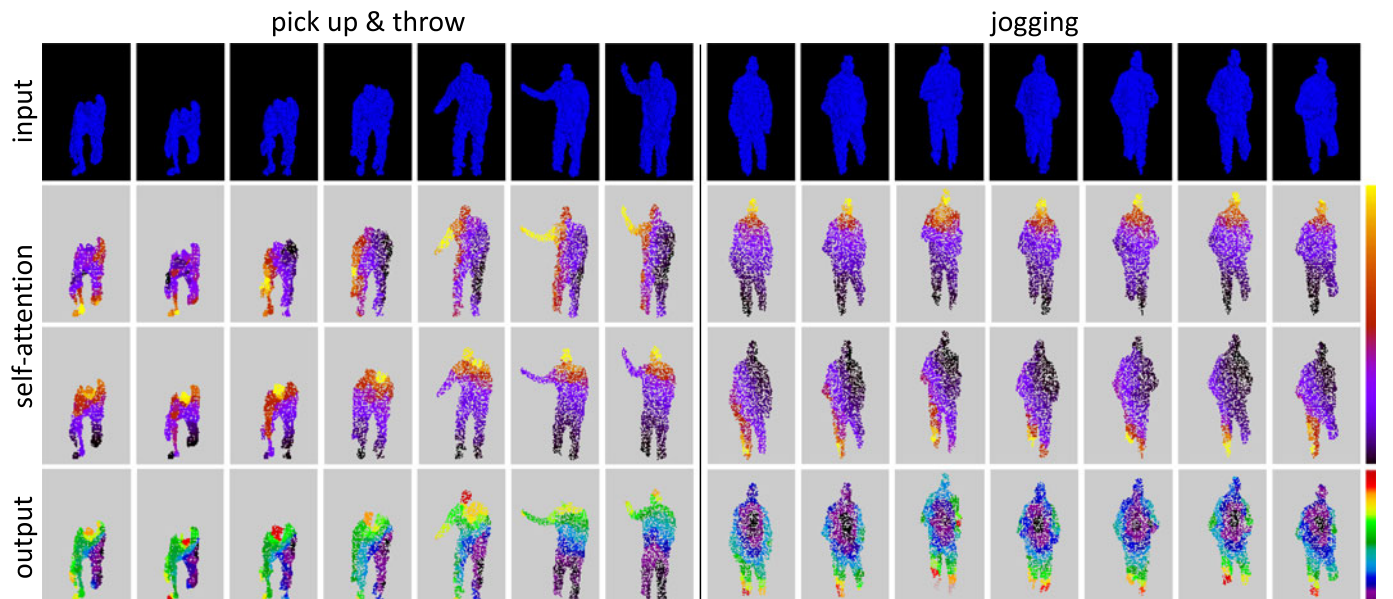


Fig. 6. Visualization of the attention and output of PST-Transformer. Brighter color indicates higher attention or activation. Our PST-Transformer pays correct attention across the frames. Moreover, PST-Transformer outputs high activations to salient moving areas, demonstrating the ability to capture the most informative clues in action reasoning.

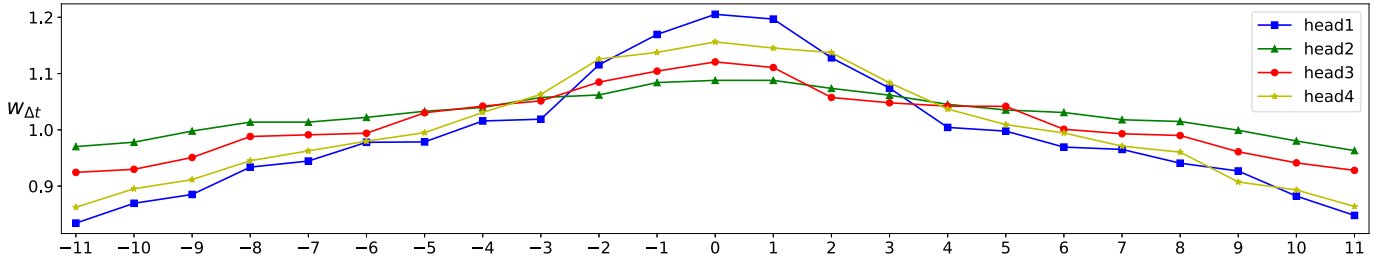


Fig. 7. Visualization of the learned parameters of temporal functions $T_{\Delta t}$. Closer frames affect more on the current moment. Moreover, parameters appear to follow Gaussian distributions with different standard deviations, like different heads capturing the temporal structure at different scales.

contains 4 heads and 2 layers. We train our model for 75 epochs with the SGD optimizer. Batch size is set to 8. Learning rate is set to 0.01, and decays with a rate of 0.1 at the 30th, 40th and 50th epochs, respectively. The mean Intersection over Union (mIoU) is used as the evaluation metric. Point colors are used.

As seen in Table 5, the proposed PST-Transformer with 3 frames outperforms the state-of-the-art methods. The Synthia 4D dataset contains 12 categories. Our PST-Transformer achieves eight best accuracies among them. We also observe that tracking does not always improve accuracy. Specifically, the MeteorNet-*m* model with point tracking obtains a slightly lower accuracy than without tracking. We speculate that this phenomenon might be caused by unreliable point tracking results, which hinder temporal modeling.

Although segmentation can be done on a single frame, exploring temporal correlation would help understand the structure of scenes and thus improve segmentation accuracy and robustness to noise, especially on the categories that are not easy to be segmented with static point clouds. For example, probably because sidewalks are close to roads, it is not easy to recognize them based on a single frame. Because exploring temporal correlation can increase their discriminativeness, the accuracy is improved. For other categories, e.g., vegetation, using a single frame already achieve the high accuracy. In this case, exploiting temporal information does not significantly improve the accuracy.

We visualize two segmentation results from the Synthia 4D dataset in Fig. 8. Our method can accurately segment most objects and make fewer incorrect predictions than other methods.

4.3 Ablation Study

Video-level self-attention, spatio-temporal encoding as well as spatio-temporal decomposition are important designs in our PST-Transformer. In this section, we perform 3D action recognition on NTU RGB+D 60 with the cross-subject protocol to investigate their impacts on point cloud video modeling. We also investigate the influence of the depth (the number of layers) and the number of heads in this section.

4.3.1 Spatio-Temporal Encoding

As shown in Table 6, integrating spatio-temporal encoding effectively improves accuracy. Compared to the baseline that does not encode spatio-temporal structure, spatial modeling leads to an improvement of 1.5%. Based on the spatial encoding, adding the temporal encoding in a decomposition manner further increases the accuracy by 1.7%.

Note that, compared to the baseline that is only based on point features, P4Transformer integrates 4D positions into both self-attention (queries and keys) and values. Because point positions also reflect point relationships to an extent, integrating 4D positions into queries and keys can help self-attention, especially when point features are not discriminative enough. Moreover, for 3D action recognition, the positions of body components may have a certain pattern, e.g., human heads are usually at high positions, integrating point positions into values can improve the discriminativeness of point features. Therefore, P4Transformer significantly outperforms the baseline without spatio-temporal encoding. We also integrate 4D position into the self-attention of PST-Transformer, which does not lead to an improvement, as shown in Table 7. The reason is that,

TABLE 5
4D Semantic Segmentation Results (mIoU %) on the Synthia 4D Dataset [5]

Method	Input	# Frames	Track	Bldn	Road	Sdwlk	Fence	Vegitn	Pole	Car	T. Sign	Pedstrn	Bicycl	Lane	T. Light	mIoU
3D MinkNet14 [5]	voxel	1	-	89.39	97.68	69.43	86.52	98.11	97.26	93.50	79.45	92.27	0.00	44.61	66.69	76.24
4D MinkNet14 [5]	voxel	3	-	90.13	98.26	73.47	87.19	99.10	97.50	94.01	79.04	92.62	0.00	50.01	68.14	77.46
PointNet++ [34]	point	1	-	96.88	97.72	86.20	92.75	97.12	97.09	90.85	66.87	78.64	0.00	72.93	75.17	79.35
MeteorNet- <i>m</i> [27]	point	2	✓	98.22	97.79	90.98	93.18	98.31	97.45	94.30	76.35	81.05	0.00	74.09	75.92	81.47
MeteorNet- <i>m</i> [27]	point	2	✗	97.65	97.83	90.03	94.06	97.41	97.79	94.15	82.01	79.14	0.00	72.59	77.92	81.72
MeteorNet- <i>l</i> [27]	point	3	✗	98.10	97.72	88.65	94.00	97.98	97.65	93.83	84.07	80.90	0.00	71.14	77.60	81.80
PSTNet [12]	point	1	-	96.32	98.07	85.40	94.66	97.16	97.51	94.83	76.65	76.99	0.00	75.39	76.45	80.79
PSTNet [12]	point	3	✗	96.91	98.33	90.83	95.00	96.96	97.61	95.15	77.45	85.68	0.00	75.71	77.28	82.24
P4Transformer [11]	point	1	-	96.76	98.23	92.11	95.23	98.62	97.77	95.46	80.75	85.48	0.00	74.28	74.22	82.41
P4Transformer [11]	point	3	✗	96.73	98.35	94.03	95.23	98.28	98.01	95.60	81.54	85.18	0.00	75.95	79.07	83.16
PST-Transformer	point	1	-	94.46	98.13	89.37	95.84	99.06	98.10	96.00	80.41	87.58	0.00	75.25	80.84	82.92
PST-Transformer	point	3	✗	96.10	98.44	94.94	96.58	98.98	98.10	96.06	82.67	87.86	0.00	76.01	81.67	83.95

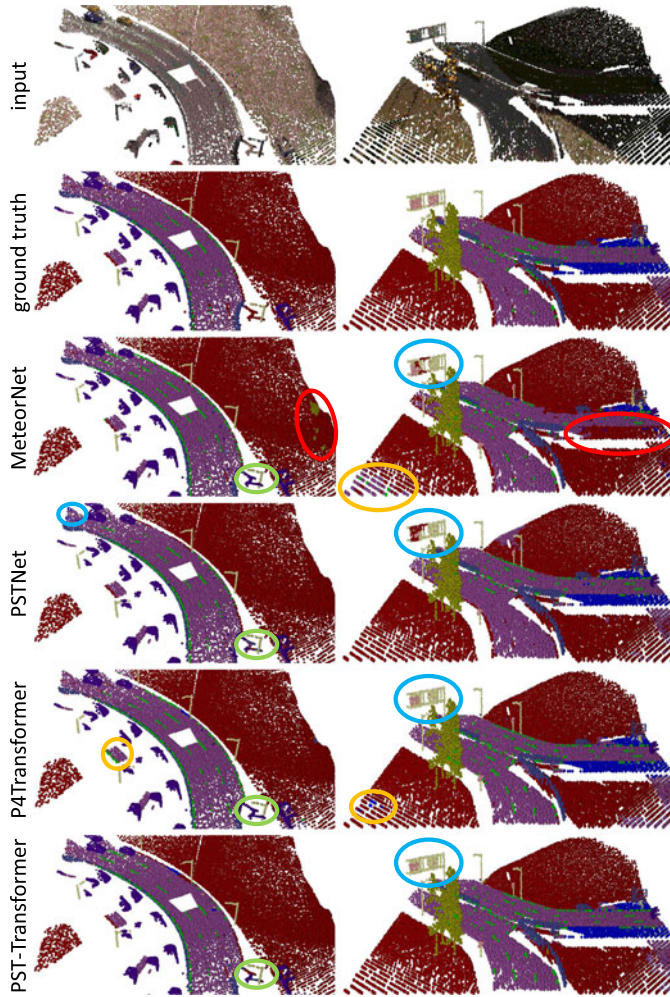


Fig. 8. 4D semantic segmentation visualization on the Synthia 4D [5] dataset. Incorrect predictions are marked with circles. All the methods achieve satisfactory results. However, our PST-Transformer makes fewer incorrect predictions than other methods.

benefiting from spatio-temporal structure encoding, point features are discriminative enough. Moreover, even for two extremely close points, they may belong to different human or object components. Additionally employing 4D positions for self-attention is not conducive to structure preservation. In summary, embedding 4D positions can help self-attention when point features are not very discriminative. Once point features are strong enough, 4D positions are not necessary and may disturb self-attention and structure preservation.

TABLE 6
Influence of Spatio-Temporal Encoding and Decomposition on NTU RGB+D 60 [37] (Cross-Subject)

Spatial	Temporal	Decompose	Accuracy (%)
×	×	×	87.8
✓	×	×	89.3
×	✓	×	86.4
✓	✓	×	90.4
✓	✓	✓	91.0

TABLE 7
Comparison of P4Transformer and PST-Transformer

Method	Query and key	Value	Accuracy (%)
Baseline	×	×	87.8
P4Transformer [11]	position	position	90.2
PST-Transformer (ous)	×	structure	91.0
	position	structure	90.6

P4Transformer Integrates 4D Position Encoding Into Both Self-Attention (Queries and Keys) and Values While PST-Transformer Only Integrates Spatio-Temporal Structure Encoding Into Values. Experiments are Conducted on NTU RGB+D [37] With the Cross-Subject Protocol

4.3.2 Spatio-Temporal Decomposition

To investigate the spatio-temporal modeling without decomposition, we treat point cloud videos as 4D point sets by integrating timestamps into coordinates. As shown in Table 6, compared to the method with only spatial modeling, which obtains an accuracy of 89.3%, adding the temporal modeling without decomposition increases the accuracy by 1.1%. Based on the improvement, the decoupled method further increases the accuracy by 0.6%, showing the effectiveness of decoupled spatio-temporal structure encoding.

4.3.3 Video-Level Self-Attention

To demonstrate the effectiveness of video-level self-attention, we also employ frame-level self-attention on NTU RGB+D 60 [37] with the cross-subject evaluation protocol, which achieves 84.2%. In contrast, video-level self-attention achieves 91.0%, which is much better than frame-level self-attention. This verifies the intuition that the frame-level self-attention, which assumes that each point appears in all frames, is not suitable for point cloud video modeling because points may flow in and out across frames, especially for long videos.

4.3.4 Depth and Multi-Head Attention

Like most deep neural networks, using multiple layers can increase the learning ability of our model. As shown in Fig. 9a, with more layers, PST-Transformer can achieve better accuracy. However, an excessive number of layers decrease the accuracy. This is because, when the network become deeper, too many parameter are involved, making the network difficult to train.

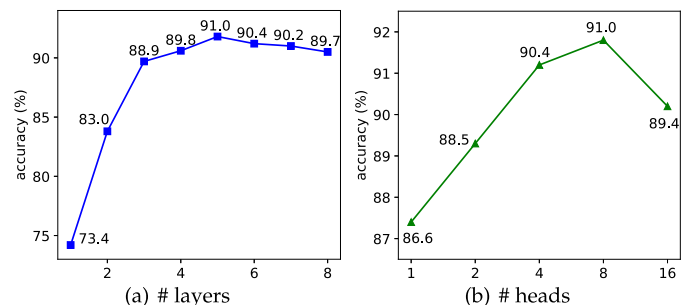


Fig. 9. Influence of the number of PST-Transformer layers and the number of heads on NTU RGB+D 60 [37] with the cross-subject evaluation protocol. Note that, when we change the number of heads, the total feature dimension of each layer is fixed.

To investigate the influence of the number of heads on our PST-Transformer, we keep the total feature dimension fixed (1024). Suppose there are h heads, then the feature dimension of each self-attention is $1024/h$. As shown in Fig. 9b, more heads can effectively increase accuracy. However, using too many heads makes the feature of each head too short and fail to carry enough information for performing attention. As a result, the accuracy decreases.

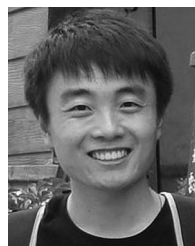
5 CONCLUSION

In this paper, we propose a Point Spatio-Temporal Transformer (PST-Transformer), which is able to preserve spatio-temporal structure of raw point cloud videos by adaptively searching related or similar local areas across entire videos. Moreover, to learn informative representations from raw point cloud videos, we integrate the ability of spatio-temporal structure encoding into PST-Transformer. Finally, because point coordinates are irregular and unordered but point timestamps exhibit regularities and order, we decouple the spatio-temporal encoding in the proposed PST-Transformer to reduce the impact of the spatial irregularity on the temporal modeling. By incorporating the proposed PST-Transformer into deep networks, our method is competent to address various point cloud video tasks. Extensive experiments on 3D action recognition and 4D semantic segmentation demonstrate that PST-Transformer effectively models point cloud videos.

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